# Mahmoud Alyosify

**NLP Engineer**

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## SUMMARY

NLP engineer working on **search, question answering, fine-tuning and evaluation** of language models. Built document question answering with hybrid search and re-ranking (top-1 accuracy from 75% to 94%), fine-tuned Llama-3-8B on 159,826 records, and wrote a master's thesis on shorter LLM answers that stay complete (47.6% fewer tokens, tested in seven languages). MSc in Artificial Intelligence, Queen's University, Canada (GPA 4.30 / 4.30).

## SKILLS

**NLP:** Transformers, RAG, semantic and hybrid search, re-ranking, question answering, information extraction, text classification

**Models:** fine-tuning (LoRA, QLoRA), Llama, Mistral, Qwen, sentence-transformers

**Tools:** Hugging Face, PyTorch, LangChain, LangGraph, FAISS, NLTK

**Evaluation:** LLM-as-judge, BERTScore, ROUGE, NLI-based content coverage, retrieval metrics (top-1, MRR), multilingual testing

**Programming:** Python, SQL, FastAPI, Docker, Git

## EXPERIENCE

**AI Engineer Intern** — Oct 2026 – Present
Injaz Tech (Injaz Group) — Maadi, Cairo

- Built question answering over company documents with hybrid search (BM25 + embeddings), **re-ranking** and query rewriting; top-1 retrieval accuracy rose from **75% to 94%**.
- Designed the evaluation: retrieval benchmarks (top-1, MRR), category-level scores and an LLM judge checked against a fixed metric (agreement 0.82).
- Covered the service with **136 automated tests** (pytest) and an evaluation check that compares each new version with the last.

**Part-Time Machine Learning Instructor** — Jul 2025 – Present
National Telecommunication Institute (NTI) — Egypt

- Teach **90-hour machine learning bootcamps** and AI for Business modules in NTI's Summer Internships Program.

**Systems Engineer (Military Service)** — Feb 2024 – Mar 2025
National Company for Roads Building & Development — Cairo

- Supported the backend systems of Egypt's national toll network, used by **more than half of the country's vehicles** across 70%+ of national roads.

**Online Instructor** — 2020 – Present
Udemy & YouTube (Arabic courses) — Remote

- Created Arabic courses on algorithms, discrete mathematics, probability and statistics: **10,000+ learners, 700K+ views**.

## PROJECTS

**MinimalLM — Master's thesis** — 2026
*Python, PyTorch, Hugging Face Transformers, NVIDIA A40*

- A training-free method that makes open LLMs (Mistral-7B, Llama-3.1-8B, Qwen2.5-3B) give shorter answers that are still complete: **47.6% fewer tokens, 19–68% faster, no answer cut off** (0 of 990).
- Tested in Arabic, Chinese, French, German, Spanish and Russian with the same settings: **45.6% fewer tokens**, 0.6% of answers cut off.
- Answer quality held: LLM-judge score 8.04/10 against 8.42 for the unchanged models (cutting the originals to the same length scores 6.46); the saving holds in six other languages (45.6%).

**Horus-OSINT — fine-tuned LLM on AWS** — 2026
*Llama-3-8B, QLoRA, Hugging Face, AWS*

- Fine-tuned **Meta-Llama-3-8B** on **159,826** global event and terrorism records (GTD, GDELT) and deployed it on **AWS** as an assistant that answers questions in plain language.

- Turned the raw GTD and GDELT records into a training set and fine-tuned the model with **4-bit QLoRA**.

**HORUS Sentinel — multi-agent intelligence platform** 2026
*Python, LLM agents, knowledge graph, self-hosted LLM*

- Turns continuously collected text into a knowledge graph of entities and relations, which a self-hosted LLM uses to write sourced reports.

**Automated Multimodal Agent — PDF to narrated slides**
*Python, LLM APIs, text-to-speech*

- An agent that turns a PDF into a narrated PowerPoint deck: it reads the document, plans and writes the slides, and records the narration.

**Human vs. AI-Agent Pull Requests — research study**
*Python, GitHub data mining, statistics*

- Built a reproducible pipeline that mines GitHub to compare review effort on pull requests written by AI agents and by people.

Also: AudioShield (deepfake audio detection) · SimCLR-Vision-SSL (84.30% top-1, self-supervised) · SCRAP (satellite collision risk, 48+ hours ahead).

## EDUCATION

**MSc in Artificial Intelligence** Sep 2025 – Oct 2026
Queen's University Kingston, Canada

- GPA **4.30 / 4.30** · Completed October 2026 · **Digilians Presidential Scholarship** (competitive national AI fellowship)
- Thesis: MinimalLM — shorter LLM answers that stay complete (supervisor: Dr. Rahatara Ferdousi)
- Coursework: Large Language Models, Generative AI, Deep Learning, Reinforcement Learning, Cloud Computing, End-to-End ML Pipelines

**BSc in Computer & Information Science (Bioinformatics)** Sep 2019 – Jul 2023
Assiut University Assiut, Egypt

- GPA 3.53 / 4.00 (Very Good with Honours) · Graduation project graded A+
- Graduation project: Vitalism Solution, vital signs from ordinary video; **Microsoft Imagine Cup 2023 semi-finalist**

## CERTIFICATIONS

- IBM AI Developer Professional Certificate (IBM, 2026)
- Generative AI & Prompt Engineering (IBM)
- Deep Learning Specialization (DeepLearning.AI, 2026)
- Machine Learning Specialization (DeepLearning.AI)
- AWS Academy: Machine Learning Foundations, ML for NLP, Data Engineering

## AWARDS & LEADERSHIP

- 1st Place, Smart Cities Hackathon 2022 (77 university teams)
- Semi-Finalist, Microsoft Imagine Cup 2023
- Finalist, ISEIC 2023 (international research competition)
- 1st Place, Science & Technology Content Competition (university level)

## LANGUAGES

Arabic (native) · English (professional working proficiency)